

AdeptiaFileTransferUtility

Version 1.0.0

Client Integration Guide

Adeptia Custom Plugin — FTP / FTPS / SFTP File Routing

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1. Overview

AdeptiaFileTransferUtility is a custom Java plugin for the Adeptia integration platform that automates file transfer between FTP, FTPS, and SFTP servers. It supports two modes:

Mode	Description
Routed Transfer	Extracts a Group ID from each filename, looks it up in a Multi Value Map XML file, and delivers the file to the correct sub-directory on the target server. Ideal for use cases where files must be routed to different folders based on content.
Direct Transfer	Moves or copies all files from a source directory to a single target directory . Ideal for simple file move/copy use cases.

Key Capabilities

- Supports FTP, Explicit FTPS, Implicit FTPS, and SFTP (including Azure Blob SFTP)
- Two transfer modes: Routed (with XML mapping) and Direct (no mapping required)
- Extracts Group ID from filename pattern: prefix_GroupId_suffix.ext (routed mode only)
- Sequential or parallel transfer mode with configurable concurrency
- Auto-creates target directories if they do not exist
- Handles file conflicts by appending a timestamp to the filename
- Supports Adeptia Key Manager for SSH private key authentication
- Auto-detects and converts PuTTY PPK keys to OpenSSH format
- Move mode (delete source after transfer) or Copy mode (keep source)
- Reports per-file success/failure back to the Adeptia process flow context

2. Prerequisites

Adeptia Platform

- Adeptia Connect 4.x or later
- Sys Admin access required to deploy JAR into Shared/Ext/

Adeptia Activities Required

- FTP Account activity for the source server
- FTP Account activity for the target server
- (Optional) Key Manager activity — required for SFTP with private key authentication
- Custom Plugin activity to invoke the JAR
- (Routed mode only) Multi Value Map activity to define Group ID to folder mapping

3. Implementation Steps

Step 1 — Deploy the JAR File

Deploy the plugin JAR to the Shared/Ext directory. Refer to the Adeptia documentation for deploying custom libraries:

[Adeptia Docs: Custom Libraries](#)

Step 2 — Restart Adeptia Microservices

Once the JAR is uploaded, restart the Runtime and Webrunner microservices to make the JAR available to process flows.

Step 3 — Create a Multi Value Map (Routed Mode Only)

Skip this step if you are using Direct Transfer mode (no XML mapping needed).

For Routed Transfer, create a Multi Value Map activity in Adeptia. Refer to:

[Adeptia Docs: Value Map](#)

Once created, Adeptia generates an XML file at:

```
/shared/web/applets/mapping/multiValuedMap/global/<Multi Value Map Name>.xml
```

This XML file path is passed as the xmlPath parameter in your Custom Plugin script.

Step 4 — Get the FTP Account Entity ID

To get the FTP Account Entity ID, open the FTP Account in Adeptia and copy the Entity ID shown on the activity details page.

View FTP: Test_Dummy_New	
Description	Test_Dummy_New
Owner's Group	administrators
Entity Id	1485626149170913280
Type Id	FTPAccount:1485626149170913280
Creation Date	03/23/2026 18:37:56
Last Modified Date	03/23/2026 18:56:39
Last Modified By	admin
Permissions	Owner(R,W,X) Group(R,W,X)
Host Name	ftp2.adeptia.com
Port	22
FTP Type	SFTP
User Id	devteam
Password	*****
Transfer Type	PASSIVE
Transfer Mode	Binary

4. Direct Transfer Mode (No XML Mapping)

Use Direct Transfer when you simply want to move or copy all files from a source directory to a target directory without any routing logic. No Multi Value Map or XML file is required.

| *New in v1.1.0 — this mode was added to support clients who do not need Group ID routing.*

Method Signatures

Two overloaded methods are available for Direct Transfer:

Option A — Minimal (4 parameters). Uses sequential mode and move semantics by default:

```
List summary = AdeptiaFileTransferUtility.transfer(
    srcAccountId,    // Source FTP Account Activity ID
    tgtAccountId,    // Target FTP Account Activity ID
    "/source/",      // Source directory path
    "/target/"       // Target directory path
);
```

Option B — Full control (7 parameters):

```
List summary = AdeptiaFileTransferUtility.transfer(
    srcAccountId,    // Source FTP Account Activity ID
    tgtAccountId,    // Target FTP Account Activity ID
    "/source/",      // Source directory path
    "/target/",      // Target directory path
    false,           // parallelMode: false=sequential, true=parallel
    5,               // concurrencyLimit: max concurrent transfers (parallel only)
    true             // deleteSourceFile: true=MOVE, false=COPY
);
```

Behaviour

- All files found in srcPath are transferred to tgtPath as-is — filenames are preserved
- No Group ID extraction or XML lookup is performed
- Target directory is created automatically if it does not exist
- File conflicts are handled by appending a timestamp to the filename
- The return format and all aft_* context variables are identical to Routed Transfer

| *Example: File report.csv in /source/ is transferred to /target/report.csv — no sub-directory is appended.*

5. Routed Transfer Mode (With XML Mapping)

Use Routed Transfer when files must be delivered to different sub-directories based on a Group ID extracted from the filename. This requires a Multi Value Map activity and the generated XML file path.

Group ID Extraction

The plugin extracts the Group ID from the second underscore-delimited segment of the filename:

```
Pattern: prefix_GroupId_suffix.ext
Example: EENav_101_Transaction.csv -> Group ID = "101"
Example: SIC_202_Report.txt -> Group ID = "202"
```

Routing Result

Source Filename	Group ID	Target Path
EENav_101_Txn.csv	101	/target/EENav_Custom1/EENav_101_Txn.csv
EENav_102_Txn.csv	102	/target/EENav_Custom2/EENav_102_Txn.csv
EENav_999_Unknown.csv	999 (no match)	/target/EENav_Base/EENav_999_Unknown.csv (default)

6. Setting Up the Custom Plugin Activity

Create a Custom Plugin activity in Adeptia and configure it with the appropriate BeanShell script from Section 4 or Section 5.

1. In Adeptia Designer go to Services > Process Flow > Create New
2. Give the process flow a descriptive name (e.g. SIC_FTP_FileRouter)
3. Create a Custom Plugin activity and use the script from the relevant section above
4. Update all account IDs, paths and parameters with your actual values
5. Refer to the screenshot below for the Custom Plugin configuration

Description* CP_SFTP_File_Transfer

acter Set Encoding* UTF-8 Refresh

Variables

#	Name	Type	Default Value
1	SrcAccountId	Plain Text	1314607238266417
2	TgtAccountId	Plain Text	1314607238266417
3	SourcePath	Plain Text	/SFTP/Saurabh_SS
4	TargetBasePath	Plain Text	/SFTP/Saurabh_SS

Number of Rows at Position Add Row Remove Row

Interpreter Type* ☒ Script ☐ Custom API ☐ Spring Bean

6. Save the activity and add it to your process flow
7. Add downstream activities as needed (e.g. alerts if `aft_failed > 0`)

7. Parameter Reference

Routed Transfer — Full Method (8 parameters)

Parameter	Type	Default	Description
SrcAccountId	String	(required)	FTP Account Activity ID for the source server
TgtAccountId	String	(required)	FTP Account Activity ID for the target server
SourcePath	String	(required)	Full path to the source directory
TargetBasePath	String	(required)	Base path on target. Sub-directories appended from routing table
XmlPath	String	(required)	Absolute path to Multi Value Map XML file on the Adeptia server
parallelMode	boolean	false	false = sequential. true = parallel
concurrencyLimit	int	5	Max concurrent transfers (parallel mode only). Renamed from maxThreads in v1.1.0
deleteSourceFile	boolean	true	true = MOVE (delete after transfer). false = COPY (keep source)

Direct Transfer — Minimal Method (4 parameters)

Parameter	Type	Default	Description
srcAccountId	String	(required)	FTP Account Activity ID for the source server
tgtAccountId	String	(required)	FTP Account Activity ID for the target server
srcPath	String	(required)	Full path to the source directory
tgtPath	String	(required)	Full path to the target directory — files land here directly

Direct Transfer defaults to: parallelMode=false, concurrencyLimit=1, deleteSourceFile=true. Use the 7-parameter overload to override these.

Context Variables Set by the Plugin

Context Key	Description
TotalFileReceived	Total number of files found in the source directory
TotalSuccessFiles	Number of files transferred successfully
TotalFailedFiles	Number of files that failed to transfer
FailedFileList	List of files which fail to process
CurrentFTPActiveConnection	Set only if connections were not properly closed — indicates an error condition

8. Sequential vs Parallel Mode

	Sequential (parallelMode=false)	Parallel (parallelMode=true)
Connections	1 source + 1 target shared across all files	1 source + 1 target per concurrent file
Speed	Slower — files processed one at a time	Faster — multiple files simultaneously
Server load	Low — single connection	Higher — multiple concurrent connections
Best for	Servers with connection limits, small batches	Large batches on servers allowing concurrent connections
Connections	2 (1 src + 1 tgt)	2 x concurrencyLimit